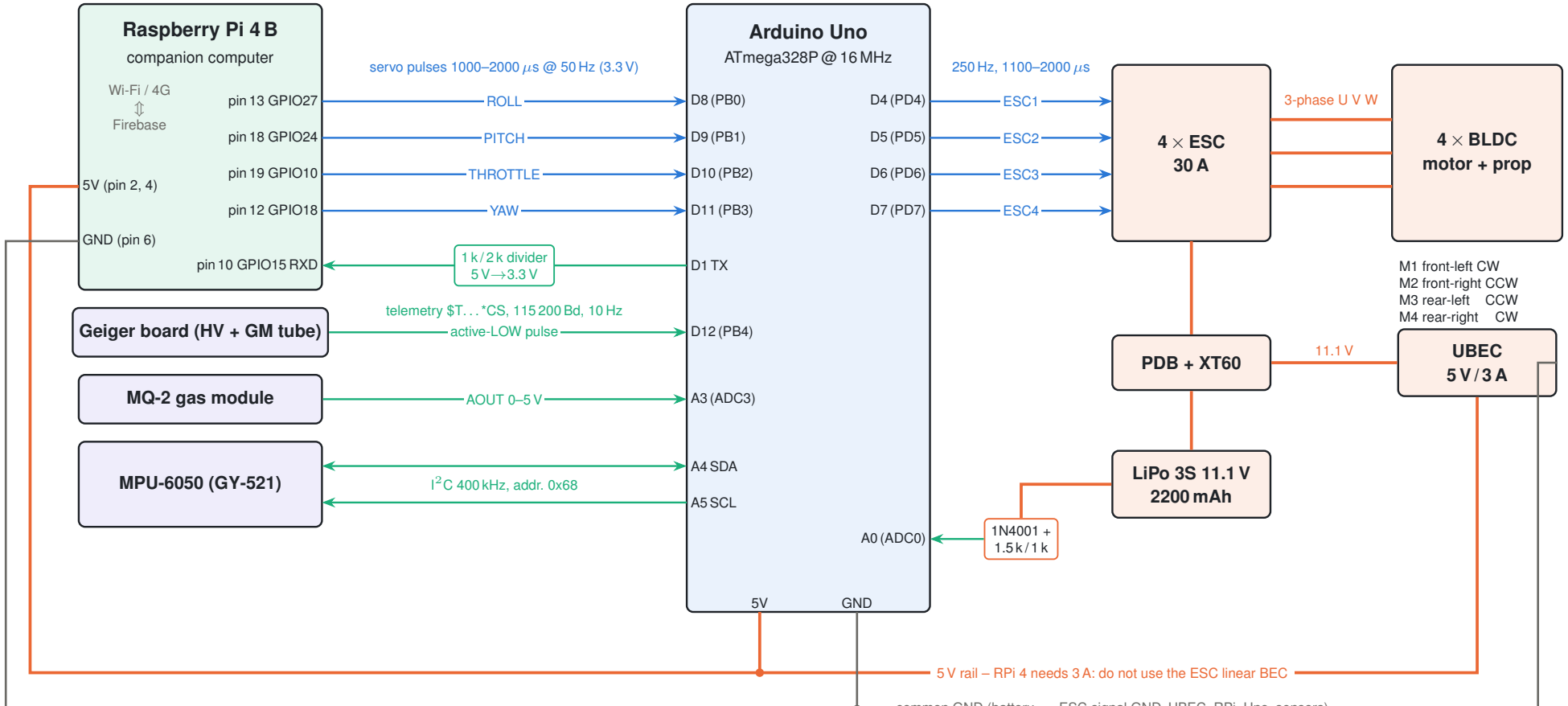


Sensors are supplied from the Uno 5 V pin (MQ-2 heater ≈150 mA). Pin numbers = physical header pins of the Pi (BCM numbers in brackets in the software).
RPi→Uno: a 3.3 V HIGH is above the ATmega $V_{IH} = 0.6\,V_{CC} = 3.0$ V, so no level shifter is needed (small margin; a 74HCT125 buffer is the robust option).
Uno TX→RPi RX **must** go through the divider: $V = 5\,\text{V} \cdot \frac{2\,\text{k}}{1\,\text{k}+2\,\text{k}} = 3.33\,\text{V}$.